

SK Saif Ibna Ezhar Arko

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 [LinkedIn](#) —  [GitHub](#) —  [Portfolio](#)

Professional Summary

AI/ML Engineer and Software Developer experienced in building production-grade Python systems, end-to-end machine learning pipelines, scalable data services, and production AI infrastructure. Hands-on with TensorFlow, Keras, PyTorch, and scikit-learn across NLP, time-series forecasting, classification, computer vision, and LLM inference. Experienced in self-hosted LLM deployment, agentic AI, multi-provider LLM routing, and AI workflow automation. Comfortable across the full model lifecycle—data ingestion, feature engineering, training, experimentation, evaluation, deployment on Linux, serving models via REST APIs, and monitoring. Strong foundation in software engineering, data structures, algorithms, competitive programming, SQL, distributed systems, and MLOps.

Technical Skills

Programming Languages:	Python, C, C++, JavaScript, SQL, PostgreSQL, MySQL
Programming & Problem Solving:	Data Structures, Algorithms, Competitive Programming, Algorithmic Problem Solving, Debugging, LeetCode, Codeforces
Artificial Intelligence:	Artificial Intelligence, LLMs, Agentic AI, AI Agents, Intelligent Automation, LLM Inference, Inference Engineering, LangGraph, Multi-Provider LLM Routing, AI Workflow Automation
ML & Deep Learning:	PyTorch, TensorFlow, Keras, scikit-learn, Transformers, LSTM/BiLSTM, CNN, Transfer Learning, Random Forest, Gradient Boosting, Logistic Regression, ARIMA
Data Engineering:	ETL Pipelines, Data Ingestion, Web Scraping, Selenium, BeautifulSoup, Requests, Pandas, NumPy, Exploratory Data Analysis, Data Cleaning, Feature Engineering
Model Development:	Model Development, Model Evaluation & Experimentation, Cross-Validation, Hyperparameter Tuning, Class Imbalance Handling, Precision/Recall/F1, ROC-AUC, RMSE/MAE, Model Performance Optimization
Web Frameworks & APIs:	Django, Django REST Framework, FastAPI, Flask, nginx, REST APIs
DevOps & Infrastructure:	Docker, Git, Jupyter, Unit Testing, Linux, llama.cpp, llama-server, systemd, Production LLM Inference, GGUF
AI Platforms & Providers:	NVIDIA NIM, DeepSeek, OpenRouter, Ollama, Transformers, LangGraph
Core Competencies:	Distributed Systems, Data Analysis, Fraud Detection, Intelligent Automation, Software Engineering, MLOps

Professional Experience

Datasoft Systems Bangladesh Limited — *AI Software Developer* *December 2025 – Present*

- Engineer production-grade Artificial Intelligence and Python-based solutions, including self-hosted LLM inference, agentic AI workflows, and scalable AI services for enterprise and financial application workloads.
- Deploy and operate self-hosted LLM inference infrastructure with llama.cpp/llama-server, serving a 228B-parameter (10B active) Mixture-of-Experts model in quantized GGUF format across heterogeneous single-GPU nodes, sized for 10–20 bursty concurrent users.
- Designed the Hermes Agent framework for model switching and multi-provider LLM routing across NVIDIA NIM, DeepSeek, OpenRouter, and local Ollama, exposed through an authenticated API gateway running as a supervised systemd service on a shared multi-user server.
- Built LangGraph-based agentic SQA workflows that plan and execute multi-step verification tasks across Django/React/Post projects, translating engineering domain knowledge into reusable AI automation.
- Delivered iSAGE, an internal AI assistant UI over the local LLM backend, integrating a bearer-authenticated inference REST API and a modular skills layer for task-specific agent capabilities.
- Designed and maintained distributed web scraping pipelines and reliable ETL workflows supporting large-scale data collection, plus reproducible ingestion and preprocessing frameworks for downstream ML/DL models.
- Deploy and operate Python services on Linux, supporting production workloads with attention to reliability, scalability, and latency.
- Collaborate in a fast-paced environment using Agile/Scrum methodologies, Git version control, and rigorous code reviews to maintain production-quality Python code.

Maxcode Lab — *Python Developer Intern* *October 2025 – December 2025*

- Developed and optimized Python automation tools for structured web scraping, data extraction, and ETL pipeline development.
- Authored clean, maintainable scripts to extract, filter, validate, and store unstructured data from diverse online sources.

- Implemented database integrations and REST API connectors for seamless data flow into analytic pipelines.
- Wrote unit tests and validation routines to ensure data quality and support continuous integration and reliable pipeline operation.

Caretutors Technology Limited — *Verified Python/CS Tutor*

December 2024 – Present

- Provide technical mentoring in Python programming, data structures, algorithms, and software engineering principles.
- Designed project-based learning modules covering data preprocessing, model training, evaluation workflows, and reproducible experiments to accelerate student readiness for ML engineering roles.

Machine Learning Projects

Text Similarity Classifier — BiLSTM Siamese Network

[\[GitHub\]](#)

- Engineered an end-to-end text similarity matching system using Bidirectional LSTM Siamese networks in TensorFlow/Keras, applicable to search relevance, ranking, and record matching.
- Developed a hybrid feature pipeline combining deep-learning embeddings with classical NLP metrics including Jaccard similarity and fuzzy string matching to improve matching precision.
- Implemented automated data augmentation to generate 7,400+ synthetic training samples and applied class weighting to address dataset imbalance.
- Optimized training with early stopping and checkpointing; evaluated models using accuracy, precision, recall, and F1 to guide model selection.

Stock Price Forecasting — ARIMA vs. LSTM

[\[GitHub\]](#)

- Implemented and compared statistical and deep-learning approaches for time-series forecasting of Microsoft stock prices with a focus on generalization to unseen periods.
- Handled end-to-end time-series preprocessing including stationarity checks, differencing, scaling, windowing, and leakage-free train/test splitting.
- Evaluated models on held-out data using standard forecasting error metrics and residual analysis to inform model selection.

Binary Classification Benchmark — Logistic Regression, Random Forest, Gradient Boosting

[\[GitHub\]](#)

- Benchmarked three classifiers on two real-world prediction tasks (Titanic survival and Bank Marketing deposit conversion) to demonstrate applicability to churn, conversion, and fraud-detection scenarios.
- Built the full machine learning pipeline including EDA, missing-value handling, categorical encoding, feature engineering, and cross-validation.
- Compared models on accuracy, precision, recall, and ROC-AUC with performance visualizations to support model selection.

Medical Image Classification — EfficientNetB3 Transfer Learning

[\[GitHub\]](#)

- Built a production-grade computer-vision pipeline using EfficientNetB3 transfer learning on a limited, imbalanced image dataset.
- Implemented targeted regularization, data augmentation, and hyperparameter tuning to reduce overfitting and maximize classification accuracy.
- Developed reusable preprocessing and evaluation workflows, including confusion-matrix and per-class metric reporting.

SmartCare: Healthcare Management Web Application

[\[GitHub\]](#)

- Architected a full-stack web application using Django with relational database design and RESTful API endpoints.
- Implemented secure data handling, user authentication, patient portals, and administrative dashboards to support healthcare workflows.

Data Collection & Automation Tools

[\[GitHub\]](#)

- Built a headless Selenium tool to validate large batches of video and playlist URLs and export structured results to Excel/CSV for downstream analysis.
- Implemented an API-based collection pipeline using the YouTube Data API to search, paginate, and assemble structured datasets for model training and analytics.

Education

BRAC University — *Bachelor of Science in Computer Science and Engineering*

Graduated: 2026

CGPA: **3.20 / 4.00**

Notre Dame College, Dhaka — *Higher Secondary Certificate*

2018-2020

GPA: **5.00 / 5.00**

St. Gregory's High School & College, Dhaka — *Secondary School Certificate*

2018

GPA: **5.00 / 5.00**

Languages

Bengali (Native), English (Full Professional Proficiency)

References

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